

9/3/2026
Engineering

IMM Framework and Production process

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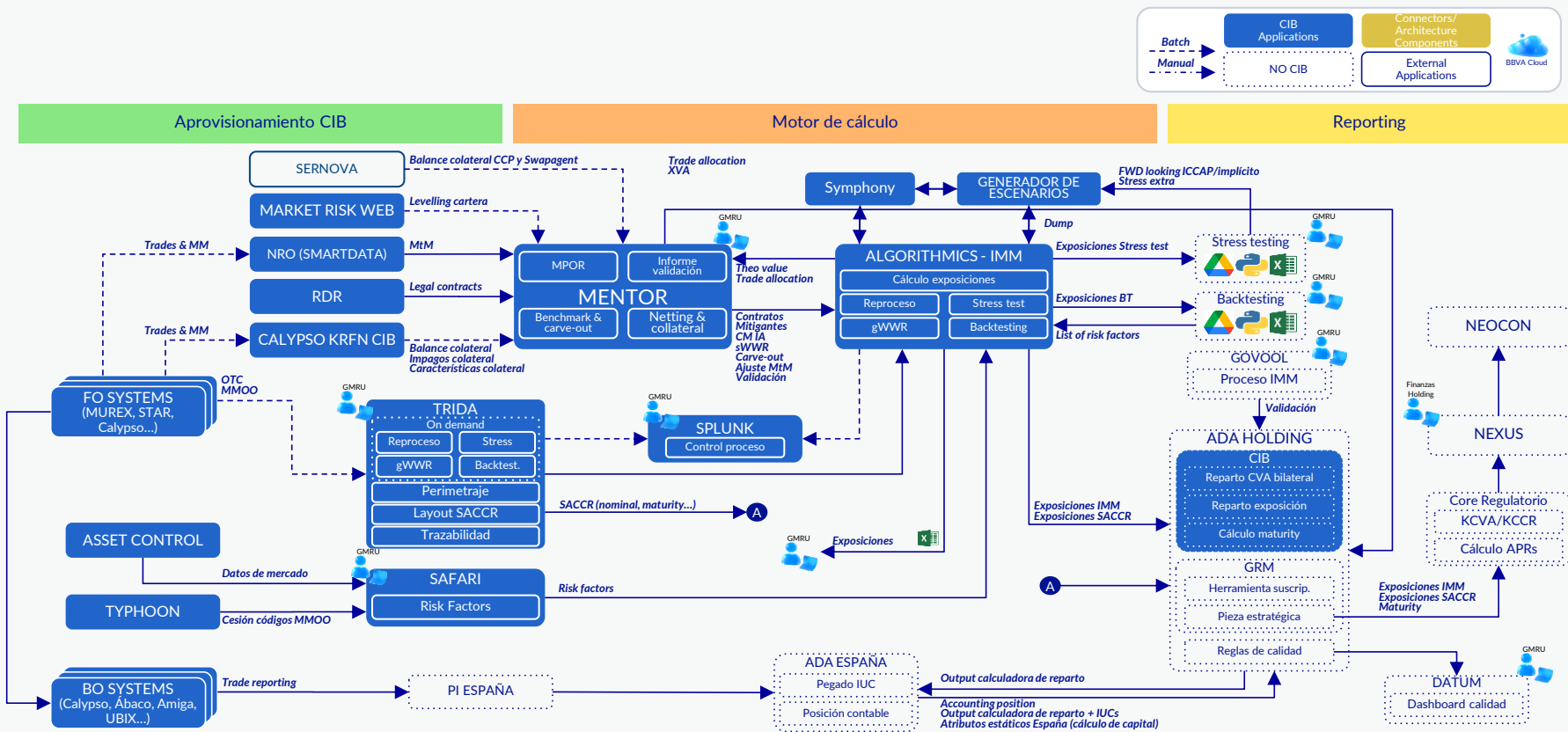
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Overview



Key Systems

		System	Infrastructure	User Interface
CIB Provisioning	SAFARI	Market Data - SAFARI: Acts as the "Golden Source" for market data and prepares the Risk Factor List (RFL)	ETL In house development based in Java Oracle DB. Load source systems data with Sql Loader, process with business logic and extract in CSV format	Java web based interface with struts framework and jsp
	NTRIDA	Trade Perimeter - NTRIDA: Centralizes trades from FO (Murex, Calypso, STAR). Applies exclusion rules (IMM vs. SA-CCR).	ETL In house development based in Java Oracle DB. Oracle DB. Load source systems data with Sql Loader, process with business logic and extract in CSV format	Java web based interface with struts framework and jsp
	Mentor	Contractual Master Data - Mentor: Manages Netting/CSA rules and collateral. Performs MtM benchmarking	In house development based in Java and Oracle DB. Batch File loaders and Online messages, process with business logic and extract in CSV format Web application for management and reporting.	Java web based interface with spring and hibernate framework

Key Systems

		System	Infrastructure	User Interface
Calculation Engine	YASE	Scenario Generation - YASE: Simulation scenario generator	In-house development using a Windows .NET library (C#, VB, C++) Computing Machine - GRID + Symphony	Does not exist
	Algorithmics	Simulation Core - ALGORITHMICS: Simulation-based portfolio valuation and metrics aggregation engine	SS&C Proprietary software (C++, JAVA, perl, python, shell) Oracle DB Computing Machine - GRID + Symphony / AWS	Does not exist
	ADA CIB	ADA: Allocates the results of the exposure calculation by trades, CVA, Maturity	In-house cloud platform. Cloud provider AWS.	ADA Console (in-house web interface) Databricks Workspaces (Jupyter Notebooks)

Key Systems

		System	Infrastructure	User Interface
Calc. Engine	Regulatory Core	Regulatory core: Credit APR's calculation engine, including CCR and CVA	In-house development on the APX infrastructure using Java-based software Databases: APX, files, and MongoDB, with monthly batch process outputs sent to the ADA environment for loading	Does not exist

Key Systems

		System	Infrastructure	User Interface
Reporting	NEXUS	NEXUS is responsible for generating some of NEOCON's input data in a centralized and standardized manner.	In house development of an Spark Scala process over ADA infrastructure (AWS)	Batch process without user interface España: Pablo Antonio Sanz Rincón. Holding: Ignacio Muñoz Maroto.
	NEOCON	The COREP C34 reports are generated within Neocon for official ECB submission ensures full data traceability. This process includes a data validation step against financial statements and final conversion to XBRL format via XWand software.	In house development based in Visual Basic .NET An Oracle DB with 48 CPUs using stored procedures, with Webfocus serving as the works-reporting software of end-users. For official EBA and BdE reporting we use Fujitsu XWand interstage	Windows Forms interface



Corporate &
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Functional Description

Systems Overview

		Description	IMM/SA-CCR process	Frequency
Front Office	Murex	BBVA Global Markets Front-to-Risk tool, utilized by Front Office desks for trade booking and capture, as well as for trade valuation, portfolio simulation, risk assumptions, and portfolio management activities.	Delivery of variation, inventory, and exoticity files, as well as the representative portfolio to generate Backtesting exposures. Delivery of market data to SAFARI for backtesting. Delivery of collateralized trades to Calypso. Delivery of trades to Mentor for MtM comparison with Algorithmics	Business days + weekly
	STAR	Internal Front Office system providing functionalities for Pre-Deal activities, Trade Booking (manual or automated capture), product pricing, and various portfolio management activities.	Sends Fixed Income and FX variation and inventory files to NTRiDA (ETL), and collateralized trades to Calypso.	Business days
Collaterals	Calypso	Back-office system containing all collateral balance information.	Supplier of collateral data	Business days
	SERNOVA	Global platform for the management, administration, and valuation of collateral and guarantees linked to Global Markets (CIB) operations. It acts as the central repository for the legal agreements that enable counterparty risk mitigation.	Supplies collateral data. It is responsible for valued collateral balances and the agreement parameters required for EPE calculation	Business days
Market data	Asset Control	Provider system for price capture and market data.	Source of market data for Algorithmics.	Business days
	TYPHOON	Internal market data server (real-time and end-of-day, calibrated or uncalibrated) for all CIB asset classes.	Supplies futures codes to SAFARI and integrates with IPV Manager to consolidate prices based on logical rules.	Business days

Systems Overview

		Description	IMM/SA-CCR process	Frequency
ETL	NTRiDA	The Bank's ETL system, responsible for trade mapping and translation between the Algorithmics Risk systems and Front Office systems (Murex, Star, Calypso). It adapts Front Office trade data to Risk modeling requirements.	Trade grouping and attribute mapping for calculation purposes. It generates data extracts with the defined scope for IMM and SA-CCR.	Business days + weekly
	SAFARI	The Bank's ETL system, responsible for mapping and translating market data between the Algorithmics Risk systems and the provider (Asset Control).	Market Data provider and mapping/translation for calculations.	Business days + weekly
Repository	NRO	Repository containing MtM valuation information from Front Office systems.	Supplies Mentor with trade valuations provided by Front Office systems to carry out the Benchmarking process.	Business days + weekly

Flowchart Production Process

On Demand - Stress - Monthly process

FORWARD LOOKING ICAAP

Initials Conditions

Scenario calculation

Exposure Calculation
Forward Looking ICAAP

Results interpretation

SCENARIO EVENTS - ICAAP impact template provided by the Research team,
INPUTS - Used during the IMM Counterparty Risk Measurement process

YASE - retrieves historical inputs and adjusts the risk factors to the ICAAP scenario

ALGORITHMICS

Market Data - Impact with ICAAP

Exposure Calculation with ICAAP Scenario and new market Data

COE - STRESS - The GMRU user performs a comparison of the HS-ICAAP metrics.

FORWARD LOOKING IMPLICIT

Initials Conditions

Scenario calculation

Exposure Calculation
Forward Looking Implicit

Results interpretation

SCENARIO EVENTS - GMRU executes the ESPS-FL tool to generate the impacted initial conditions
NPUTS - Used during the IMM Counterparty Risk Measurement process

YASE - It retrieves historical inputs and adjusts the risk factors to the Implicit scenario

ALGORITHMICS

Market Data - Impact with Implicit

Exposure Calculation with Implicit Scenario and new market Data

COE -STRESS - The GMRU user performs a comparison of the HS-Implicit metrics

Systems Overview

		Description	IMM/SA-CCR process	Frequency
Mentor	Consolidation and structuring of regulatory inputs	Integrates the methodological tagging of products (IMM/SA-CCR), counterparty hierarchies and netting sets, collateral and movements agreements, as well as penalty parameters (concentration, settlement failures, levelling, sWWR), constituting the reference structural dataset for the model	IMM markings and no-IMM marking. MPoR Penalization.	Daily
	Scope definition and delivery to the engine	The validated structural information is transmitted to Algorithmics, determining the effective IMM scope, exposure aggregation, and treatment of mitigants.	Mitigants and Hierarchy	Daily
	Benchmarking and carve-out governance	Receives the theoretical values from the Front Office and the engine, executes systematic benchmarking and, according to predefined rules and expert validation, defines IMM→SA-CCR carve-outs and MtM adjustments, which are sent back to the engine as a formal scope control mechanism	Benchmarking and carving out. MtM adjustment	Weekly + End of month
	Exposure validation	Once received the final outputs (EE, EPE, EEPE), Mentor executes Data Quality rules, node-by-node functional validations, and monthly variation analysis, culminating in the formal confirmation of figures by the responsible user.	Figure validation	Weekly + End of month
	Reprocessing module	In the event of incidents, it allows for the controlled regeneration of hierarchies, mitigants, and regulatory flags, triggering traceable and auditable recalculations	Reprocess	End of month

Systems Overview

		Description	IMM/SA-CCR process	Frequency
Algorithmics	YASE	Simulation scenario generation system	Provider of all scenarios used for both Batch and On-Demand calculations.	Weekly + End of month
	Flag I	MtM valuation system for trades within the full-valuation scope.	Executes trade filtering alongside TRIDA, generating the trade allocation for the calculation. Calculates the valuation of all trades initially within the IMM perimeter to generate the Theo-Value report.	Weekly + End of month
	Flag III	Scenario simulation portfolio valuation and metrics aggregation system.	Valuation of SA-CCR EAD exposures. Calculation of exposure figures for IMM trades. Scope profiling based on Benchmark results. Calculation of metrics at different aggregation levels according to Synthetic Netting Sets.	Weekly + End of month

Systems Overview

		Description	IMM/SA-CCR process	Frequency
Algorithmics	Reprocessing	System for rerunning Flag I and Flag III calculations.	In the event that relevant errors are found in the Flag I and Flag II processes, the user is allowed to make changes to the calculation inputs: • Trades(NTRIDA) • Static data (Mentor)	Monthly (OnDemand)
	OnDemand	System for launching additional regulatory control calculations in addition to the main process	Allows for the individual execution, with a maximum monthly frequency, of all additional processes: • Backtesting Real & Test • Backtesting Risk factors • gWWR • Stress Testing • Forward Looking ICAAP • Forward Looking Implicit	Monthly (OnDemand)

Systems Overview

	Description	IMM/SA-CCR process	Frequency
ADA	Allocates the results of the exposure calculation by trades, CVA, Maturity	Allocation of the exposures and CVA (OTC) at Front operation level Calculate the effective maturity at SNS level and operation level when there is no SNS. For the IMM operations allocates at CP level. Allocation of the exposures an CVA (MMOO) Generate the output of the output of the allocation calculator and validate the results.	Monthly (OnDemand)

Systems Overview

	Description	IMM/SA-CCR Process	Frequency
Regulatory Core	1. Regulatory CORE is a single-instance batch application that runs automatically on a monthly basis within an Engineering environment known as APX. 1. The engine has been developed using a Java framework adapted to the APX environment, and its scheduling — executed in parallel and individually by country — is triggered between the 5th and the 12th business day of each month. 1. RWA credit calculations take place in Regulatory CORE at contract level considering input data, the implemented calculation algorithms, and the intervention of parametric tables that drives key calculation steps (e.g. CCF allocation).. 1. The outputs of the Regulatory CORE engine comprise the input data, the intermediate calculations, and the final results. 1. Outputs are subsequently loaded into ADA (AWS) tables within the Regulatory CORE application, so they are made available in a sandbox for the organization..	With the IMM input, a new provision is received and processed by the CORE engine. It contains information on operations subject to the IMM perimeter, primarily highlighting exposure (EAD) and maturity, data necessary for calculating RWA.	Monthly

Systems Overview

		Description	IMM/SA-CCR Process	Frequency
Reporting	Nexus	NEXUS is responsible for generating some of NEOCON's input data in a centralized and standardized manner. It processes information from the Group's banking entities, excluding Turkey, acting as a key component within the regulatory reporting process.	NEXUS collects data from CORE Engine and adapt it into the format required by NEOCON. Additionally, it performs validation, cleansing and reconciliation processes to ensure consistency and reliability.	Monthly (OnDemand)
	NEOCON	NEOCON is the tool responsible for the generation of: Consolidated Financial Statements: Balance, Income Statement y Out of Balance.	Consolidated Regulatory Reporting: including the generation of the C34 COREP Reports for IMM Positions.	Monthly (OnDemand)

Manual Adjustment

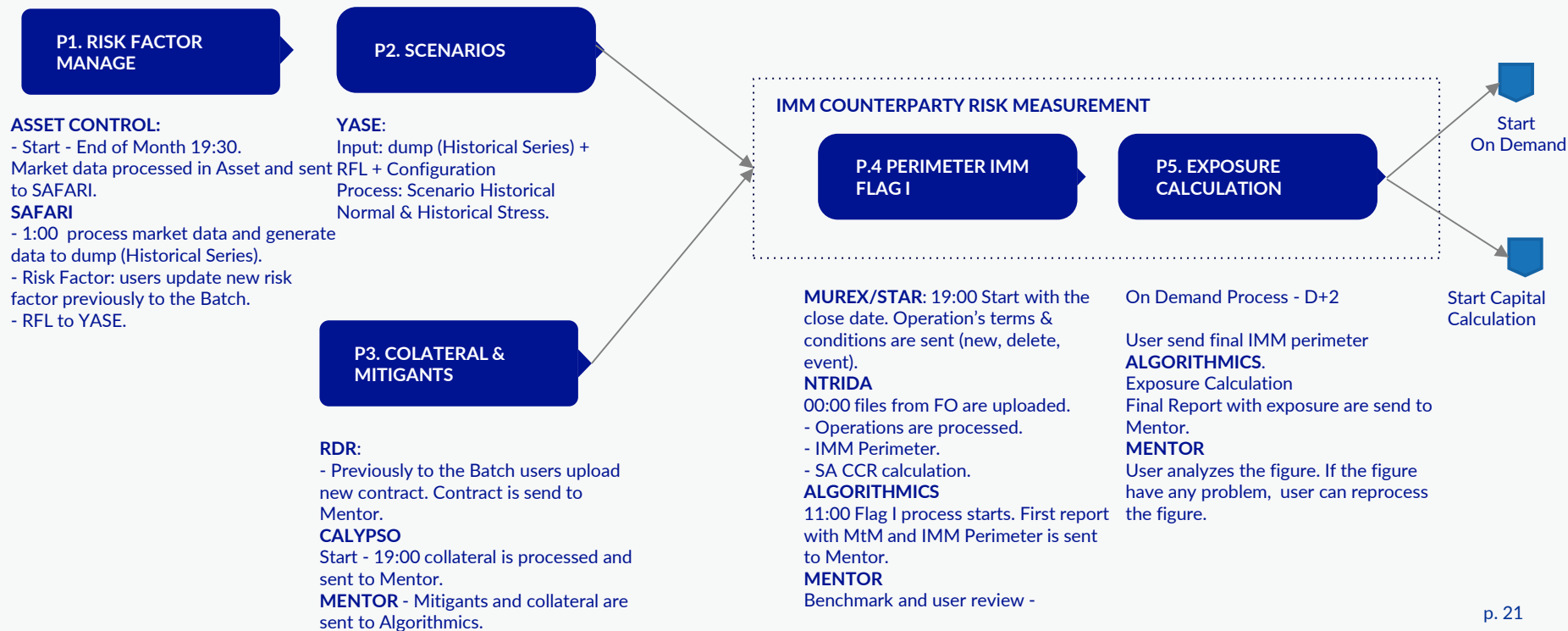
Manual adjustments are not permitted within the IMM scope:

- In the event that a trade lacks a model or does not behave as expected, its exposure is calculated using **SA-CCR SNS**.
- It does not exist sensitivity approximations.
- It is only permitted one exception. At the collateral level for Independent Amounts that are not registered in the systems.

Flowchart production process

Flowchart Production Process

IMM Counterparty Risk Measurement



Flowchart Production Process

Capital Calculation

CAPITAL CALCULATION

Allocation calculator

Data consolidation in ADA

Exposure allocation, CVA, and Maturity Calculation: Allocation of exposure figures (Flag 3) and CVA at the trade level, and calculation of M (Effective Maturity).

Capital calculator

Consolidation of information from ADA, including information already sent to SA-CCR and the output of the distribution calculator.

Regulatory Capital Calculation (RWA, KCCR, KCVA) and generation of outputs to Nexus.

REPORTING

Nexus receives information from the CORE Engine and validates data and applies the quality rules defined by Finance Holding.

To generate COREP reporting, Nexus adapts information into the format required by NEOCON. Subsequently, NEOCON consolidates the information in order to generate the official models required by the Supervisor.

Flowchart Production Process

On Demand - Stress

STRESS

Initials Conditions

Scenario Calculation

Exposure calculation with
Lehman, Covid & Stress
Extra

Results interpretation

On Demand Process - NTRiDA web.

INPUTS - Used during the IMM Counterparty Risk Measurement process

SCENARIO EVENTS - historical events Lehman and Covid and Stress Extra (most punitive year for the bank)

YASE - Generate Lehman, Covid and Stress Extra scenario.

ALGORITHMICS - calculate new metrics with Lehman, Covid an Stress extra Scenario

COE - STRESS - The results obtained for each scenario (Lehman, COVID-19, and Stress Extra) are contrasted by GMRU against the metrics generated during the standard batch (Flag 3, HN and HS scenarios)

Flowchart Production Process

On Demand - GWWR (Monthly process)

GWWR

Initials Conditions

Scenario calculation

Exposure Calculation
GWWR

Results interpretation

INPUTS - Used during the IMM Counterparty Risk Measurement process

SCENARIO EVENTS - new RFL with credit scalar.

YASE - New scenario GWWR

ALGORITHMICS - For the calculation of General Wrong-Way Risk (gWWR), Algorithmics focuses on the stochastic simulation of Probabilities of Default (PDs) through single-factor models and Fast Fourier Transform (FFT). This is followed by the adjustment of Expected Exposures (EEs) to incorporate the impact of adverse correlation risk (gWWR)

GMRU Users - The results obtained across the different scenarios are analyzed by GMRU

Flowchart Production Process

On Demand - Backtesting

REAL PORTFOLIO AND REPRESENTATIVE PORTFOLIO

Initials Conditions

IMPUTS - extracts the historized information from the On-Demand process pool, containing counterparty risk measurement data under IMM (Flag 3) for T-13 and T.

Scenario calculation

YASE - Generation of HN and HS scenarios using T-13 initial conditions and T-13 historical data

Calculation report - TAU y EPE Media

ALGORITHMICS - Real Portfolio - 13-month historical portfolio. The calculations are performed on this specific portfolio

Representative Portfolio - A fictitious portfolio provided by MX based on technical portfolios

Results interpretation

RAG TOOL - Reports for both the real and representative portfolios are processed in the Traffic Light Tool, which classifies model performance into traffic-light-coded zones.

FACTOR DE RIESGO

Initials Conditions

IMPUTS - Algorithmics extracts the historized information from the On-Demand process pool, containing counterparty risk measurement data under IMM (Flag 3) for T-13 and T

Scenario calculation

YASE - A new BT risk factor scenario is generated, based on a reduced universe of risk factors (RF).

Cálculo informes (TAU y EPE Media)

ALGORITHMICS - Deviations are analyzed using key metrics such as TAU at 95%. The results are consolidated into reports, which Algorithmics sends to the Traffic Light Tool

Results interpretation

RAG TOOL - Reports for both the real and representative portfolios are processed in the Traffic Light Tool, which classifies model performance into traffic-light-coded zones.

Reconciliation Process

Traceability IMM Process

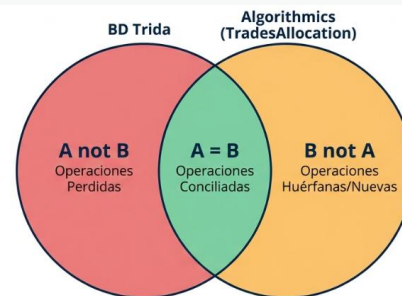
The main objective is to perform **trade-level tracking from the Front Office (FO)** systems to the final calculation reports generated by the risk engine for Internal Model Method (IMM) figures. This procedure ensures traceability throughout the **IMM Counterparty Calculation process**.

Phase 2 - Reconciliation model

Traceability consists of cross-referencing trades between the Trida input from FO and the final Algorithmics reports.

Three main possibilities may arise for the trades:

- **PAIRED (A=B):** Trades found in both Trida and the Algorithmics reports.
- **LOST (A not B):** Trades present in Trida but missing from the final Algorithmics report.
- **NEW (B not A):** Trades that appear in Algorithmics but are not found in Trida



NEW:
Manual adjustments of
Independent Amount
Directly injected MtM
adjustments

LOST

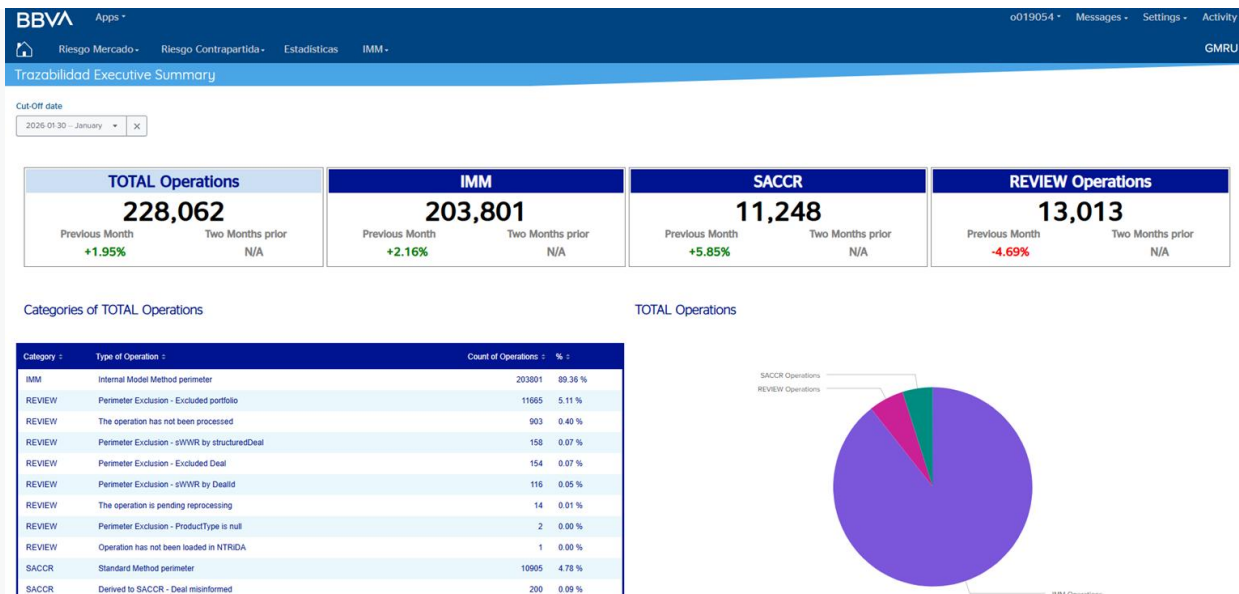
NTRiDA errors -loading failure, processing error, extraction failure.

Algorithmics errors - database (DB) loading failure.

Exclusiones normativas: trades excluded due to Specific Wrong-Way Risk (sWWR).

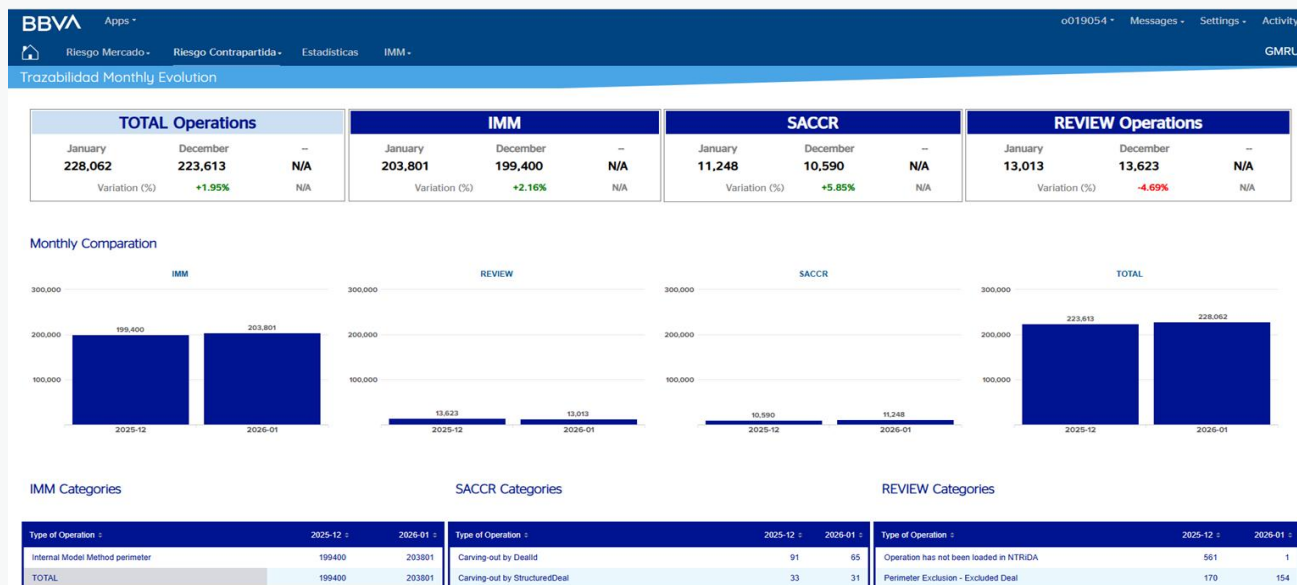
Traceability IMM Generation

The output of the traceability process is a report generated within the **Splunk tool**. In this tool, users can visualize information at three levels: **execution summary**, **SA-CCR traceability**, and a **comparison of results from the last three months**.



Traceability IMM Generation

The output of the traceability process is a report generated within the **Splunk tool**. In this tool, users can visualize information at three levels: **execution summary**, **SA-CCR traceability**, and a **comparison of results from the last three months**



DATA QUALITY IMM

Architecture and approach

This process ensures execution integrity and data quality within the IMM Counterparty Calculation and On-Demand processes.

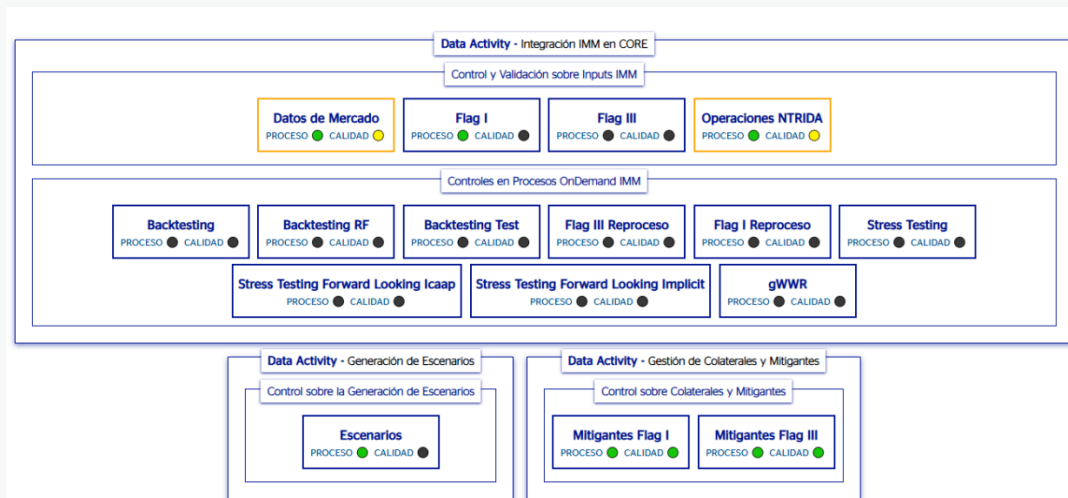
- **Operational Pillar (Splunk):** Used to actively manage the quality of IMM processes, review rule execution statistics, and perform technical error tracing.
- **Functional Pillar (Datum):** A corporate governance tool that documents the control framework and architecture, defining responsibilities across functional areas and ensuring regulatory compliance.

Dataactivity	Responsible
Exposure calculation, on-demand processes, and integration of exposures with CORE.	Counterparty Exposure Metrics
Collateral and Mitigants Management.	Counterparty Exposure Metrics
Scenario Generation.	Admission and CCR Measurement

DATA QUALITY IMM

Monitoring

Splunk - Responsible users come here to intuitively view the status of processes. These are presented in coloured boxes (blue for pending, green for completed correctly, yellow for warnings, and red for errors) and indicate specific control statuses (e.g. failures in empty input files, layout controls, etc.).



- **Blue** - Process pending execution.
- **Green** - Process completed successfully.
- **Yellow** - Process completed successfully with some warnings.
- **Red** - Erroneous Process.

DATA QUALITY IMM

Functional validation

Datum - After the technical process, the owners complete the "Data Activities." This is formalized through mandatory self-declarations and month-end questionnaires, where they certify whether the execution was successful and if the inputs were received in a timely and proper manner. Furthermore, they formally validate compliance with critical rules and overall Data Quality (DQ).

The screenshot displays the BBVA Data Quality IMM interface, divided into two main sections: 'BUSINESS GLOSSARY' and 'DATA ACTIVITIES'.

BUSINESS GLOSSARY: This section shows a list of data activities. The table includes columns for 'ESTADO' (Status), 'NOMBRE' (Name), 'DESCRIPCIÓN' (Description), 'TIPOLOGÍA DQ' (DQ Typology), 'PROCESO' (Process), and 'MARCA DE CRÍTICA' (Critical Mark). Three activities are listed, all with a status of 'In progress' and a critical mark of '< 36072'.

ESTADO	NOMBRE	DESCRIPCIÓN	TIPOLOGÍA DQ	PROCESO	MARCA DE CRÍTICA
In progress	Cálculo de exposición, procesos on dem...	Cb (Risk in market unit) Market - Item	En el contexto del cálculo de KCCR regulatorio, B...	N/A	< 36072 Other Key Data Proc...
In progress	Generación de Escenarios	Cb (Risk in market unit) Market - Item	Actividad que consiste en la creación de escenari...	N/A	< 36072 Other Key Data Proc...
In progress	Gestión de Colaterales y Mitigantes	Cb (Risk in market unit) Market - Item	Actividad cuyo objetivo es integrar la gestión de ...	N/A	< 36072 Other Key Data Proc...

DATA ACTIVITIES: This section provides a detailed view of a specific data activity. The title is 'Cálculo de exposición, procesos on demand e integración de exposiciones con CORE'. It includes a description of the activity and a list of related data quality metrics. The 'INFORMACIÓN DEL DATA ACTIVITY' section is divided into three main areas: 'JERARQUÍA DE GOBIERNO' (Government Hierarchy), 'TAXONOMÍA CORPORATIVA' (Corporate Taxonomy), and 'INFORMACIÓN ADICIONAL' (Additional Information). The 'JERARQUÍA DE GOBIERNO' section shows a hierarchy of roles and responsibilities. The 'TAXONOMÍA CORPORATIVA' section shows the internal model for counterparty credit risk. The 'INFORMACIÓN ADICIONAL' section shows additional information related to the activity, including a list of critical rules and overall data quality metrics.

DATA QUALITY

Dataactivity	Process Control	TOM	System	Control
Exposure calculation, on-demand processes, and exposure integration with CORE	Control and Validation of IMM Inputs	IMM Perimeter - Carving-Out	Algorithmics	Input Existence Empty Inputs Layout Control Difference Threshold Special Character Control (o Invalid Character Control) Scenario Holiday AIDB Inputs AIDB Loading AIDB Invalid Products AIDB Excluded Strategies AIDB Reference Control
	IMM Counterparty Risk Calculation	Exposure Calculation - Validation Reports	Algorithmics	Trade Allocation Control EAD Attribution SA-CCR EAD TASD (Technical acronym, usually remains unchanged) Exposure All Nodes TRIM Final Report HN (Usually "Head Node" or internal ID) Enriched Theo-Value (Theoretical Value) Final Report All Nodes EEPE Report by Trade
		Exposure Calculation - Calculation of the final exposure	Algorithmics	Difference threshold

DATA QUALITY

Dataactivity	Process Control	TOM	System	Control
Exposure calculation, on-demand processes, and exposure integration with CORE	Controls in IMM OnDemand processes	Backtesting - Backtesting (EPE)	Algorithmics	Festive Scenarios
		Backtesting - Risks Factors	Algorithmics	Festive Scenarios
		gWWR - Data integration and processing	Algorithmics	Festive Scenarios
		Stress Testing - Forward Looking	Algorithmics	Festive Scenarios
		Stress Testing - Forward Looking Implicit	Algorithmics	Festive Scenarios
		Stress Testing - Historical Stress Periods	Algorithmics	Festive Scenarios
	General control of Allocation	Allocation calculator - Data integration and processing	ADA	Review Technical DQ Report on inputs
		Allocation calculator - Exposure Allocation, CVA and Maturity Calculation	ADA	Validate delivery engine figures

DATA QUALITY

Dataactivity	Process Control	TOM	System	Control
Scenario Generation	Control over scenario generation	Scenario Generation	YASE	Control of scenario volume with respect to previous execution Control of scenario volume with respect to historical data Control of scenario emptiness Control over the Stress date in the Extra Stress scenario Control of statistics Generic control of inputs
		Risk Factor Management – Market Data Capture and Processing	YASE	ASE Dump Integrity Check
Collateral and Mitigant Management	Collateral and Mitigants	Collateral and Mitigants - Collateral balance	Mentor	Theo Value Mentor MBBMRSTBACK1 Mentor MBBMRSTBACK2 Mentor MBBMRSTBACK3 Mentor MBBMRSTBACK4 NROskeletal NRO Star Nominal

Thank you for your time



Corporate &
Investment Banking